

Ahashan Habib Niloy

📍 Kentucky, USA

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Education

University of Kentucky

PhD in Computer Science

Kentucky, USA

Fall 2024 - Present

BRAC University

Bachelor of Science in Computer Science and Engineering; GPA [3.90/4.00]

Dhaka, Bangladesh

June, 2021

- **Selected Coursework:** Programming Language, Data Structures and Algorithms, Discrete Mathematics, Computer Networks, Operating Systems, Artificial Intelligence, Software Engineering, Elements of Statistics and Probability, Linear Algebra and Fourier Analysis

Rangpur Cadet College

Higher Secondary Certificate; GPA [5.00/5.00]

Rangpur, Bangladesh

August, 2016

- **Selected Coursework:** Information and Communication Technology, Physics, Math and Biology

Rangpur Cadet College

Secondary School Certificate; GPA [5.00/5.00]

Rangpur, Bangladesh

May, 2014

- **Selected Coursework:** Information and Communication Technology, Physics, Math and Biology

Work Experiences

Graduate Teaching Assistant

Department of Computer Science, University of Kentucky

Kentucky, USA

Fall 2025 - Present

Graduate Research Assistant

Department of Computer Science, University of Kentucky

Kentucky, USA

Fall 2024 - Summer 2025

- Area of Research: Multimodal AI, Federated Learning, Algorithm

Lecturer

Department of Computer Science and Engineering, BRAC University

Dhaka, Bangladesh

January 2023 - Present

- Courses Conducted: Programming Language I, Data Structures, Digital Logic Design, Database Systems

Adjunct Lecturer

Department of Computer Science and Engineering, BRAC University

Dhaka, Bangladesh

October 2021 - December 2022

- Courses Conducted: Digital Logic Design, Database Systems, Computer Interfacing, Introduction to Robotics

Undergraduate Teaching Assistant

Department of Computer Science and Engineering, BRAC University

Dhaka, Bangladesh

February 2020 - December 2020

- Course Assisted: Algorithm

Research Experience

Undergraduate Thesis

Department of Computer Science and Engineering, BRAC University

Dhaka, Bangladesh

May 2020 - June 2021

- Thesis Title: Comparative Study of X-ray and CT Scan Images for the Detection of COVID-19 using Deep Learning.
- Thesis Supervisor: Dr. Mohammad Zavid Parvez

Peer Reviewed Publication

Conference

Rahman M.J., **Niloy, A.H.**, Shiba, S.A., Fahim, S.M.F.A., Faria, F.N., Hussain, E., Parvez, M.D.(2021) CoroPy: A Deep Learning Based Comparison Between X-Ray and CT Scan Images in Covid-19 Detection and Classification. In: Rojas I., Castillo-Secilla D., Herrera L.J., Pomares H. (eds) Bioengineering and Biomedical Signal and Image Processing. BIOMESIP 2021. Lecture Notes in Computer Science, vol 12940. Springer, Cham.
https://doi.org/10.1007/978-3-030-88163-4_34

Journal

Niloy AH, Fahim SMFA, Parvez MZ, Shiba SA, Faria FN, Rahman MJ, Hussain E and Tamanna T (2024) CovRoot: COVID-19 detection based on chest radiology imaging techniques using deep learning. Front. Sig. Proc. 4:1384744. doi: 10.3389/frsip.2024.1384744 <https://doi.org/10.3389/frsip.2024.1384744>

Research Interests

- Multimodal AI
- Federated Learning
- Biomedical Image Processing
- Algorithms

Voluntary works

Journal Review

- Frontiers in Signal Processing - Biomedical Signal Processing
- Frontiers in Public Health - Injury Prevention and Control

Technical Skills

Programming Languages: Python, Java
Deep Learning and ML: Pytorch, TensorFlow, Keras, Scikit-learn
Data Analysis & Visualization: NumPy, Pandas, Matplotlib
Databases: SQL, MySQL
Web Development: HTML5, CSS3, Javascript.
Tools: Git, Latex

Academic Projects

- Online Book Shop management system (Design and Analysis)
- E-commerce website (HTML, CSS, JS)
- Music Rhythm LED Flashlight Circuit
- Smart Car with obstacle-detecting ability using Arduino Uno
- Micro Controller Based Data Logger system
- Home Automation System Using Arduino Nano (Simulation-Based)

Honors and Scholarship

- **Performance-Based Scholarship** (70%) at BRAC University for four consecutive years (2017-2021).
- Secured 8th Position in Dinajpur Education Board in Higher Secondary Certificate and awarded with **Talent Pool Scholarship** (2016).
- Graduated with the **Highest Distinction** from BRAC University (2021).
- Awarded **Vice Chancellor's Honor List** 8 times as recognition of decent performance in BSc in CSE.
- Awarded **Dean's Honor List** 2 times as recognition of decent performance in BSc in CSE
- Achieved The Bronze Standard of **THE DUKE OF EDINBURGH'S INTERNATIONAL AWARD** for co-curricular and extra-curricular activities (2017).

References

Dr. Mohammad Zavid Parvez

Assistant Professor

Department of Computer Science and Engineering, BRAC University.

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Dr. Md. Khalilur Rhaman

Professor

Department of Computer Science and Engineering, BRAC University.

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Najeefa Nikhat Choudhury

Lecturer

Department of Computer Science and Engineering, BRAC University.

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